

Dead Man's Cipher — ELI5 Technical Guide

English Edition • Step-by-Step Architecture Spec

DEAD MAN'S CIPHER
Hackathon Spec Document
ELI5 Edition

1. Executive Summary (ELI5 Concept)

Dead Man's Cipher is a pirate-themed cryptographic workbench designed to encrypt, digitally sign, hide, and inspect secret maritime dispatches over open networks. It allows pirate captains to send secret treasure coordinates safely without enemy warships or codebreakers intercepting them.

2. Complete Technology Stack & Tools Used

Tool / Library	Specific Module Used	Function & Purpose
Next.js 16	App Router & Turbopack	Fast React framework for component routing & page rendering.
TypeScript	Strict Type Checking	Eliminates runtime type errors across all crypto suites.
WebCrypto API	window.crypto.subtle	Browser-native Ed25519 signatures, AES-256-GCM & PBKDF2.
Steganography	Zero-Width Unicode & Canvas 2D	Hides data in text lyrics (u200B) and PNG image LSB pixels.
Web Audio API	AudioContext Synthesizer	Synthesizes realistic ship bells, clicks, sirens & chimes in JS.
Recharts	BarChart & RadarChart	Renders live frequency histograms & 5-axis security radar.
Tailwind CSS v4	Custom CSS Tokens & Utilities	Responsive dark nautical theme & clean layout spacing.

3. Feature-by-Feature Deep Dive & Logic

Detailed step-by-step breakdown of how every single feature works under the hood:

Feature A: Identity Vault (/identity)

- **WebCrypto Keypairs:** Generates Ed25519 (or ECDSA) private/public keypairs via `crypto.subtle.generateKey`.
- **8-Character Fingerprint:** Hashes the public key with SHA-256 and formats the first 8 hex characters (e.g. 8F3A2B1C).
- **Printable SVG Wax Seal:** Dynamically renders high-res SVG vector seals displaying captain name, ship, and fingerprint.
- **BLACKSPOT Key Revocation:** Allows captains to mark stolen keys as BLACKSPOT, causing inspection to immediately block messages signed with that key.

Feature B: Sealed Missive Forge (/compose)

- **PBKDF2 Key Derivation:** Derives a 256-bit encryption key from passphrase using 100,000 PBKDF2-SHA256 rounds.
- **AES-256-GCM Encryption:** Encrypts secret coordinates with random 12-byte IV and produces authenticated ciphertext.
- **Ed25519 Digital Signing:** Signs ciphertext with captain private key to guarantee sender authenticity.
- **Roll Random Missive:** 1-click generator for random pirate treasure locations, shanties, passphrases, and keys.

Feature C: Boarding Inspection (/inspect)

- **4-Gate Defense Pipeline:** Evaluates incoming dispatches across 4 mandatory security checks:
 1. Gate 1 (Format): Validates DMC1 protocol header.
 2. Gate 2 (Replay Window): Rejects dispatches older than 24.0 hours.
 3. Gate 3 (Signature): Verifies Ed25519 signature authenticity via `crypto.subtle.verify`.
 4. Gate 4 (Revocation): Rejects keys marked BLACKSPOT.
- **Audio Cues:** Plays success ship bell on pass, or alarm siren on fail.

Feature D: Steganography Lab (/stego)

- **Sea Shanty Text Stego:** Converts secret bits to invisible zero-width Unicode characters (`\u200B` for 0, `\u200C` for 1) inserted into lyrics.
- **Sail Emblem Image LSB Stego:** Reads HTML5 Canvas pixels (`getImageData`) and modifies the Least Significant Bit (LSB) of RGB channels to store secret locations inside PNG emblem images.
- **Interactive Map Picker:** Includes `NauticalMap` component so users can pick map locations directly for image steganography.

Feature E: Kraken's Trial Attack Lab (/attack-lab)

- **4 Simulated Attacks:** Forged Signatures, 72-Hour Replay Attacks, BLACKSPOT Key Dispatches, and Tampered Ciphertexts.
- **Defense Verification:** Proves that the 4-gate inspection engine automatically catches and rejects all malicious cyber attack attempts.

Feature F: Cipher Cracker & Encrypter (/cracker)

- **Cryptanalysis Engine:** Renders letter frequency histograms, Index of Coincidence (`IoC` ~ 0.0667 English vs ~0.038 random), and Chi-Squared Caesar brute-force solvers.
- **Spacious Interactive Encrypter:** Features a large plaintext textarea, key shift slider (+1 to +25), and 1-click "Encrypt & Load into Codebreaker" button with quick presets.

Feature G: Storm Warning Dashboard & Ship's Log Audit

- **Threat Dashboard (/threat):** 5-axis Radar Chart measuring Cipher Strength, Key Freshness, Sig Assurance, Anti-Replay & Stego Stealth with a 96/100 score gauge.
- **Ship's Log Audit (/audit):** Immutable append-only audit log chained via SHA-256 hashes (`currentHash = SHA256(previousHash : seq : timestamp : event : actor : details)`). Includes 1-click integrity verification.